

Rishika Mamidibathula

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EDUCATION

Columbia University

New York, NY

M.S. Data Science, GPA: 3.9/4.0

Aug 2025 – Expected Dec 2026

- Coursework: Deep Learning, Generative AI Systems, Agentic AI, High Performance Computing, Causal Inference, Statistical Inference
- Teaching Assistant for Natural Language Processing and AI for Public Policy; Data Science Institute Student Council Member

Vellore Institute of Technology

Vellore, IN

B.Tech. Computer Science and Engineering (Data Science specialisation), GPA: 4.0/4.0

Jul 2019 – May 2023

- Ranked 7th out of 200 (Top 4%); Merit Scholarship recipient (2019–2023); Program Representative (2019–2023)

- Coursework: Machine Learning, Artificial Intelligence, Natural Language Processing, Image Processing, Social Information Networks

SKILLS

Programming & Frameworks: Python, SQL, R, C++, PyTorch, TensorFlow, scikit-learn, FastAPI, NumPy, Pandas

LLM & AI Systems: RAG, LangChain, Hugging Face, ChromaDB, FAISS, Multimodal AI, Prompt Engineering, Agent Workflows

Data Engineering & Infrastructure: Databricks, PySpark, MongoDB, MySQL, BigQuery, Redis, Docker, Git, REST APIs

Cloud & MLOps: AWS, Azure DevOps, Weights & Biases, CI/CD, Experiment Tracking, Feature Engineering, Model Monitoring

WORK EXPERIENCE

Predictive Analytics Intern

Jun 2026 – Aug 2026

NYC Administration for Children's Services (ACS)

New York, NY

- Built an explainability framework for a child welfare risk model on 900K+ records, transforming black-box predictions into transparent, policy-driven decisions through configurable operating thresholds spanning 25–80% recall (0.799 ROC-AUC, 7.2× base-rate precision).
- Designed a three-level reliability audit proving explanations reproducible across evaluation sets ($\rho=0.991$), then established via noise-control testing that only 15 of 149 features carried real signal, setting a defensible limit on what the agency can claim from any feature ranking.

Software Engineer

Aug 2023 – Jul 2025

Shell

Bengaluru, IN

- Developed machine learning forecasting solutions in Databricks (PySpark, SQL) across 12 business units, reducing forecast error by 23% and enabling \$100K+ annual cost optimization decisions.
- Built production data pipelines, feature engineering workflows, and 5 Blue Prism RPA bots with logging and retry logic, cutting manual reporting effort by 85% (120+ hours/quarter) and improving SLA compliance from 92% to 99%.

Technical Analyst Intern

Jan 2023 – Jul 2023

Novartis

Hyderabad, IN

- Engineered a NLP-driven clinical trial analysis workflow for drug-level sentiment mining, automated summarization, and outcome extraction from unstructured research documents, reducing manual literature review and evidence synthesis effort by 40% per quarter.
- Built predictive and time-series forecasting pipelines on environmental operations data, integrating web-scraped external signals, feature engineering, and temporal trend analysis to support sustainability initiatives targeting a 19% annual reduction in carbon emissions.

RESEARCH EXPERIENCE

Data Science Research Assistant – Ocean Carbon Data Valuation (McKinley Group)

Sep 2026 – Present

Columbia University, Lamont-Doherty Earth Observatory

Palisades, NY

- Extending a Data Shapley framework valuing individual SOCAT surface ocean fCO₂ observations against a physical model prior rather than an empty baseline, using KNN-Shapley approximation and spatially blocked cross-validation to target where sparse, expensive measurements would most improve the ocean carbon sink reconstructions contributed annually to the Global Carbon Budget.

Computer Vision Research Assistant – Ophthalmic AI & Eye Tracking (AI4VS Lab)

Jul 2026 – Present

Columbia University, Irving Medical Center

New York, NY

- Developing gaze-augmented deep learning models for glaucoma diagnosis that fuse clinician eye-tracking with OCT report features, aligning model attention with expert reading behavior to make automated screening for irreversible vision loss auditable enough for clinical use.

AI Engineer Research Assistant – Clinical LLM & Phenotyping

Jan 2026 – Present

Columbia University, Irving Medical Center

New York, NY

- Built an extraction system with hybrid de-identification, longitudinal EHR reconstruction, and hallucination-aware validation, converting unstructured notes into 56 cardiac sarcoidosis phenotype variables and replacing months of chart review with a manuscript-ready cohort.

Machine Learning Engineer Research Assistant – Human Rights LLM Evaluation

Jan 2026 – Present

Columbia University, Graduate School of Arts and Sciences

New York, NY

- Developed a two-stage retrieval-augmented LLM framework with automated web search and chain-of-thought reasoning scoring 27 defense manufacturers across 9 human-rights dimensions, cutting analyst review effort by 80% with kappa and Krippendorff's alpha validation.

PROJECTS

Folio: Clinical Multimodal RAG

<https://github.com/rishika1099/Folio-Clinical-Multimodal-RAG> | <https://folio-health.vercel.app>

- Built a multimodal clinical platform unifying RAG, document understanding, speech transcription, and longitudinal record management across text, PDF, image, and audio, hitting 85.1% extraction micro-F1 and 100% RAG recall@1 at sub-2s latency.

KV Cache Optimization for LLM Inference

<https://github.com/rishika1099/KV-Cache-Implementation>

- Benchmarked KV-cache optimizations for Llama-2-7B including KIVI quantization, TopK sparse selection, SnapKV eviction, and MLA latent compression with custom Triton kernels, achieving 4× compression at LongBench quality parity and a 3.1× peak throughput gain.

Colon Cancer Trial Causal Analysis

<https://github.com/rishika1099/Colon-Cancer-Trial-Causal-Analysis>

- Applied ATE estimation, heterogeneous treatment-effect modeling, and mediation analysis to a randomized colorectal cancer trial, showing that conditioning on a post-treatment variable induced collider bias severe enough to reverse the estimated effect (HR 0.69 to 1.10).